

NIBRAAS NAVAS

DATA SCIENCE & MACHINE LEARNING

+91 9656535727 | nibraasnikz@gmail.com

linkedin.com/in/nibraas-navas | github.com/NibraasNavas | nibraasnavas.github.io/Portfolio/

Professional Summary

Computer Science & Engineering (Data Science) undergraduate with hands-on experience building machine learning, computer vision, and data analytics applications. I enjoy turning real-world problems into practical software, from real-time fatigue detection on edge devices to predictive models and interactive dashboards. Strong foundation in Python, SQL, machine learning, data analysis, and model evaluation, with a focus on building reliable and useful solutions.

Technical Skills

Programming:	Python, SQL, C, Dart, HTML/CSS
Data Science:	Pandas, NumPy, Exploratory Data Analysis, Statistical Inference, Feature Engineering, Data Cleaning
Machine Learning:	Scikit-learn, TensorFlow, Regression, Classification, Model Validation, SHAP, Computer Vision
AI / Computer Vision:	OpenCV, Dlib, CNNs, Facial Landmark Detection, Image Classification
Visualization:	Power BI, Tableau, Plotly, Matplotlib, Seaborn, Interactive Dashboards, KPI Reporting
Development:	FastAPI, Django, REST APIs, Streamlit, Git, GitHub, Docker, Jupyter, VS Code
Cloud / Mobile :	Firebase, Flutter, Hive NoSQL, Raspberry Pi, Edge AI

Projects

AwareDrive: Real-Time Driver Fatigue & Safety Telemetry Python · OpenCV · TensorFlow · Dlib · Raspberry Pi
Edge Computer Vision & Real-Time Safety Monitoring

- Built a custom CNN with Batch Normalization and Dropout for driver fatigue classification, achieving **92.4% classification accuracy** across multi-subject alert and fatigue image benchmarks.
- Developed real-time facial landmark tracking using Dlib and OpenCV, calculating Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and PERCLOS-based fatigue indicators.
- Optimized the model with TFLite Float16 quantization, reducing memory usage by **50%** and achieving **15 ms inference**.
- Deployed the system on Raspberry Pi at **30+ FPS**, with visual and acoustic alerts triggered in **500 ms** under variable lighting and head-pose conditions.

Recipe Wizard: Multimodal AI Culinary Assistant Flutter · Dart · Gemini API · Firebase · Hive
Multimodal AI & Mobile Application

- Built a cross-platform mobile application using Gemini's multimodal vision capabilities to identify **15+ ingredients** from live camera captures with **2.5 s generation latency**.
- Designed structured few-shot prompts and JSON schemas to produce consistent recipe outputs while handling dietary and allergen constraints.
- Implemented offline-first local caching with Hive, reducing redundant API calls by **60%**, with Firebase synchronization for cloud-backed data.
- Developed asynchronous state management and recipe bookmarking in Flutter/Dart for responsive local retrieval and cross-device access.

Education

Bachelor of Technology (B.Tech) in Computer Science & Engineering (Data Science) 2022 – 2026
APJ Abdul Kalam Technological University, Kerala, India **CGPA: 7.11/10.0**

Relevant Coursework: Machine Learning, Statistical Analysis, Database Management Systems, Big Data Processing, Computer Vision, Algorithms

Higher Secondary Education – Science & Mathematics 2020 – 2022
Aleemul Islam Higher Secondary School, Kerala, India

Certifications

- Python for Data Science** – Infosys Springboard Feb 2025
- Statistical Inference using Python** – Infosys Springboard Mar 2025
- Fundamentals of Artificial Intelligence** – IBM SkillsBuild Oct 2026